

GENERAL EDUCATION – MATHEMATICS

I. BASIC CONCEPTS

A. FUNDAMENTAL OPERATIONS ON INTEGERS

Addition and Subtraction of Integers

When adding or subtracting integers, we need to consider whether the two numbers have the same sign or different signs.

When adding or subtracting two numbers with the same sign:

1. Add the absolute values of the numbers.
2. Keep the common sign of the numbers.

When adding or subtracting two numbers with different signs:

1. Find the absolute value of the numbers.
2. Subtract the smaller absolute value from the larger absolute value.
3. Keep the original sign of the number with the larger absolute value.

Multiplication and Division of Integers

To multiply or divide two signed numbers:

1. Multiply or divide the absolute values.
2. If the signs are the same, then the sign of the result is always positive.
3. If the signs are different, then the sign of the result is negative.

Remember the following rules:

Multiplication of Integers	Division of Integers
$(+)(+) = (+)$	$\frac{(+)}{(+)} = (+)$
$(-)(-) = (-)$	$\frac{(-)}{(-)} = (+)$
$(-)(+) = (-)$	$\frac{(-)}{(+)} = (-)$
$(+)(-) = (-)$	$\frac{(+)}{(-)} = (-)$

B. PROPERTIES AND OPERATIONS ON THE SET OF INTEGERS

For integers, there are five properties of operations. These are summarized in the following table.

Integer Property	Addition	Multiplication	Subtraction	Division
Commutative Property	$x + y = y + x$	$x \times y = y \times x$	$x - y \neq y - x$	$x \div y \neq y \div x$
Associative Property	$x + (y + z) = (x + y) + z$	$x \times (y \times z) = (x \times y) \times z$	$x - (y - z) \neq (x - y) - z$	$x \div (y \div z) \neq (x \div y) \div z$
Identity Property	$x + 0 = x$ $0 + x = x$	$x \times 1 = x$ $1 \times x = x$	$x - 0 = x$ $0 - x \neq x$	$x \div 1 = x$ $1 \div x \neq x$
Closure Property	$x + y \in \mathbb{Z}$	$x \times y \in \mathbb{Z}$	$x - y \in \mathbb{Z}$	$x \div y \notin \mathbb{Z}$
Distributive Property	$x \times (y + z) = x \times y + x \times z$ $x \times (y - z) = x \times y - x \times z$			

Note: Entries in red are not a valid property.

C. DIVISIBILITY RULES

2, 4, and 8	3, 6, and 9	7	5 and 10
• 2 – last digit is even • 4 – last two digits can be divided by 4 • 8 – last three digits can be divided by 8	• 3 – sum of digits is divisible by 3 • 6 – number is even <i>and</i> divisible by 3 • 9 – sum of digits is divisible by 9	Double the last digit, then subtract the doubled last digit from the number <i>without</i> the last digit	• 5 – last digit is either 0 or 5 • 10 – last digit is 10

D. GCF AND LCM OF 2-4 NUMBERS

Greatest Common Factor (GCF)	Least Common Multiple (LCM)
• Highest number that divides two or numbers exactly • Largest of the common factors	• Smallest number that is a multiple of two or more integers

E. RATIO AND PROPORTION

Ratio	Proportion
A ratio is a comparison between two positive quantities a and b . It is expressed as either of the following: • $a:b$ • $\frac{a}{b}$ • a is to b	A proportion is a statement of equality between two ratios. It is written as either of the following: • $\frac{a}{b} = \frac{c}{d}$ • $a:b = c:d$

II. ALGEBRAIC EXPRESSIONS

A. LAWS OF EXPONENTS

Product Rule	Quotient Rule	Power Rule
$x^a \cdot x^b = x^{a+b}$	$\frac{x^a}{x^b} = x^{a-b}$	$(x^a)^b = x^{ab}$
Negative Exponent	Zero Exponent	One Exponent
$x^{-a} = \frac{1}{x^a}$	$x^0 = 1$	$x^1 = x$

B. OPERATIONS ON POLYNOMIALS

There are four main polynomial operations, each mentioned in the following table with explanation.

Operation	Explanation
Addition of Polynomials	To add polynomials, always add <i>like</i> terms (terms having the same variable and power). Adding polynomials will result in a polynomial with the same degree.
Subtraction of Polynomials	Subtracting polynomials are the same as addition, the only difference being the type of operation. The result when subtracting two polynomials will also result to the same degree.
Multiplication of Polynomials	The product of two or more polynomials will result in a polynomial with a degree higher to either polynomial (except when one of the factors is a constant polynomial).
Division of Polynomials	Dividing two polynomials may or may not result in a polynomial. To divide polynomials, long division can be done: 1. Write the polynomial in descending order of degree. 2. Check the highest power in the dividend and divisor and divide the two. 3. Using the result from Step 2, multiply it with the entire divisor. 4. Subtract, then bring down the next term. 5. Repeat Steps 2 to 4 until there are no more terms to carry down.

6. The quotient is written as a polynomial, and the remainder written in fraction form.

C. FACTORING

Algebraic expressions can be factored using many methods. The most common methods used for factoring algebraic expressions are:

- Factorization using common factors
- Factorization by regrouping terms
- Factorization using identities

Factorization using Common Factors

To factor an algebraic expression using common factors, the highest common factor of the terms is determined and grouped accordingly. In simple terms, the reverse process of expansion of an algebraic expression is its factorization.

$$ax + ay = a(x + y)$$

Factorization using Standard Identities

An equality relation which holds true for all values of variables in mathematics is known as an identity. The following are the most common identities:

Name of Identity	Identity
Square of a Binomial	$(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$
Difference of Two Squares	$a^2 - b^2 = (a + b)(a - b)$

D. TRANSLATING ENGLISH WORDS AND PHRASES INTO MATHEMATICAL PHRASES AND VICE VERSA

The table below lists some key words and phrases that are used to describe common mathematical operations.

To write algebraic expressions and equations, assign a variable to represent the unknown number. In the table below, the letter “x” is used to represent the unknown.

In translation problems, the words “sum”, “difference”, “product”, and “quotient” imply at least two parts; use parentheses when a sum or difference is multiplied. Consider the two examples below.

The sum of three times a number and five $3x + 5$
 Three times the sum of a number and five $3(x + 5)$

Operation	Key Word/Phrase	Example	Translation
Addition (+)	Plus	A number plus three	$x + 3$
	More than	Ten more than a number	$x + 10$
	The sum of	The sum of a number and five	$x + 5$
	The total of	The total of six and some number	$6 + x$
	Increased by	A number increased by two	$x + 2$
	Added to	Eleven added to a number	$x + 11$
Subtraction (–)	Minus	A number minus seven	$x - 7$
	Less than	Four less than a number	$x - 4$
	The difference of	The difference of a number and three	$x - 3$
	Less	Nine less a number	$9 - x$
	Decreased by	A number decreased by twelve	$x - 12$
	Subtracted from	Six subtracted from a number	$x - 6$
Multiplication (×)	Times	Eight times a number	$8x$
	The product of	The product of fourteen and a number	$14x$

	Twice; Double	Twice a number; Double a number	$2x$
	Multiplied by	A number multiplied by negative six	$-6x$
	Of	Three-fourths of a number	$\frac{3}{4}x$
Division ($\div$)	The quotient of	The quotient of a number and seven	$\frac{x}{7}$
	Divided by	Ten divided by a number	$\frac{10}{x}$
	The ratio of	The ratio of a number to fifteen	$\frac{x}{15}$
Powers (x^n)	The square of; Squared	The square of a number; A number squared	x^2
	The cube of; Cubed	The cube of a number. A number cubed	x^3
Equals ($=$)	Equals	Seven less than a number equals ten.	$x - 7 = 10$
	Is	Three times a number is negative six.	$3x = -6$
	Yields	Twelve added to a number yields five.	$x + 12 = 5$
	Amounts to	Nine less a number amount to twenty.	$9 - x = 20$

E. SOLVING LINEAR EQUATIONS

There are several ways in solving linear equations. These methods for finding the solution of linear equations are as follows:

Operation	Explanation
Graphical Method	To solve linear equations graphically, graph both equations in the same coordinate system and check for the point of intersection in the graph. If there is a point of intersection, then that is the solution.
Elimination Method	In this method, either add or subtract the equations to get the equation in one variable. If the coefficients of one of the variables are the same and the sign of the coefficients are the opposite, we can add the two equations to eliminate the variable. Similarly, if one of the variables are the same in value and in sign, then we can subtract the two equations to achieve the same result. In case that one variable cannot be eliminated by addition or subtraction, then multiply one or both of the equations by a constant value on both sides of the equation to obtain the equivalent system of linear equations (make sure that one of the variables have the same numerical value), then eliminate one variable by either adding or subtracting equations.
Substitution Method	In a system of linear equations with two variables in two variables, the solution can be obtained by the Substitution Method. The following are the steps in solving via Substitution Method: 1. Simplify the given equation by expanding the parenthesis. 2. Solve one of the equations for either x or y. 3. Substitute the result in Step 2 to the other equation. 4. Solve for the variable in the new equation using elementary arithmetic operations. 5. Use the result in Step 4 to find the value of the other variable using either of the two given equations.
Cross-Multiplication Method	This is the simplest method in solving linear equations in two variables. Consider the following two equations: $a_1x + b_1y + c_1 = 0$ $a_2x + b_2y + c_2 = 0$ Using cross multiplication, the values of x and y are: $x = \frac{b_1c_2 - b_2c_1}{b_2a_1 - b_1a_2} \qquad y = \frac{c_1a_2 - c_2a_1}{b_2a_1 - b_1a_2}$ Where $b_2a_1 - b_1a_2 \neq 0$. The final solution is given as

	$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{b_2a_1 - b_1a_2}$
Matrix Method	Linear equations can also be solved using the matrix method. This method is extremely helpful for solving linear problems in two or three (or even more) variables. Consider the following equations: $\begin{aligned} a_1x + a_2y + a_3z &= d_1 \\ b_1x + b_2y + b_3z &= d_2 \\ c_1x + c_2y + c_3z &= d_3 \end{aligned}$ The following equations can then be written in matrix form as $\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ In this case, the matrices A, B, and X are: $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

III. GEOMETRY

A. BASIC TERMS

Euclid's way in presenting geometry is referred to as the axiomatic method.

In geometry, there will be certain fundamental concepts that will not be defined. These are **undefined terms**. In this case, the three undefined terms are **point**, **line**, and **plane**.

Also related to the axiomatic system are **postulates**, which are statements that describe the entire system, which will not require proof.

From these, **definitions** are precise statements regarding a certain term, which will make use of several concepts.

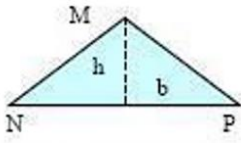
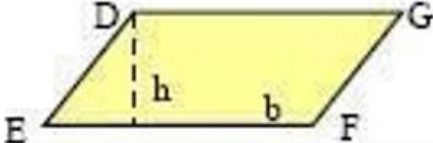
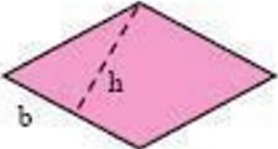
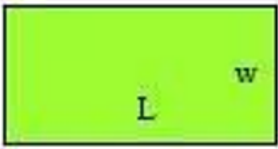
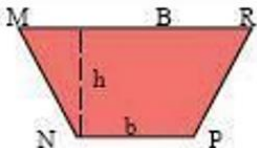
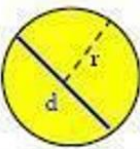
Theorems, on the other hand, are statements that are required to be proved from postulates, definitions, and previously proved theorems. From them, a **corollary** is a theorem that is an *immediate consequence* of another theorem.

Some of the most common definitions are the following:

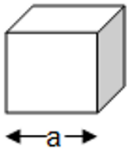
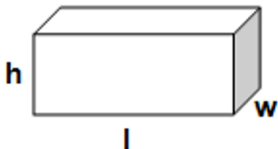
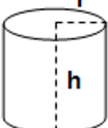
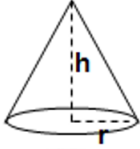
Term	Description
Intersecting lines	Two or more lines having a common point
Complementary angles	Two angles whose sum is 90 degrees
Supplementary angles	Two angles whose sum is 180 degrees
Parallel lines	Two or lines which does not share a common point
Perpendicular lines	Two lines which, when intersected, will form 90-degree angles
Angles	A figure formed from two rays with a common point (called a vertex)
Interior angle	An angle that is located <i>inside</i> a polygon
Exterior angle	An angle that is located <i>outside</i> a polygon
Perimeter	Sum of all the sides of a polygon
Area	Amount of space covered by a plane figure
Volume	Amount of space covered by a solid figure
Polygons	A closed figure made up of segments (called sides)
Triangle	A polygon with three sides
Right Triangle	A triangle with one right angle

Hypotenuse	The longest side of a right triangle
Parallelogram	A quadrilateral (4-sided polygon) with two pairs of parallel sides
Rectangle	A parallelogram with right angles
Rhombus	A parallelogram with all its sides equal
Square	A regular quadrilateral; a rhombus with right angles
Trapezoid	A quadrilateral with one pair of parallel sides
Kite	A quadrilateral with two pairs of congruent, adjacent sides
Circles	A plane figure formed with one line at a fixed distance from a different point (called a center)
Solid	A three-dimensional figure (i.e.: having length, width, and height)

B. AREA AND PERIMETER OF PLANE FIGURES

Name	Figure	Area	Perimeter Circumference (Circle)
Triangle		$A = \frac{b \times h}{2}$ <i>b</i> – base <i>h</i> – height	$P = MN + NP + PM$
Parallelogram		$A = b \times h$ <i>b</i> – base <i>h</i> – height	$P = DE + EF + FG + GD$
Rhombus		$A = b \times h$ <i>b</i> – base <i>h</i> – height	$P = 4b$
Rectangle		$A = l \times w$ <i>l</i> – length <i>w</i> – width	$P = 2l + 2w$
Square		$A = l^2$ <i>l</i> – length	$P = 4l$
Trapezoid		$A = \frac{(B + b) \times h}{2}$ <i>B</i> and <i>b</i> – bases <i>h</i> – height	$P = MN + NP + PR + RM$
Circle		$A = \pi r^2$ <i>r</i> – radius	$C = 2\pi r$

C. VOLUME OF SOLID FIGURES

Name	Figure	Volume	Variables
Cube		$V = a^3$	a – length of edge
Rectangular prism		$V = l \times w \times h$	l – length w – width h – height
Cylinder		$V = \pi \times r \times h$	r – radius of base h – height
Cone		$V = \frac{1}{3} \times \pi \times r \times h$	r – radius of base h – height from tip to base
Sphere		$V = \frac{4}{3} \times \pi \times r^3$	r – radius

IV. PROBABILITY

A. TERMS AND DEFINITIONS

An **experiment** refers to any activity, such as rolling a die.

Suppose one wants to roll a fair die. There is a total of six possibilities (1, 2, 3, 4, 5, and 6). This is referred to as the **sample space**. Formally speaking, a sample space is a set of all possible outcomes of a certain experiment.

Meanwhile, an **event** is a set of outcomes of a random experiment. From the example above, suppose we want to know how many even numbers are there in a fair die. In this case, there are three (2, 4, and 6).

Thus, the **probability** of an event occurring is expressed as:

$$P(x) = \frac{E(x)}{S(x)}$$

where $E(x)$ refers to the number of elements in an event (number of preferred outcomes) and $S(x)$ refers to the total number of elements in the sample space (total number of outcomes).

In certain concepts in probability, the concept of a factorial is sometimes needed. $x!$ (read as “ x factorial”) is defined as $1 \times 2 \times 3 \times \dots \times (n - 1) \times n$. As per definition, $0! = 1$.

B. FUNDAMENTAL COUNTING PRINCIPLES

Fundamental Principles of Counting

The Fundamental Principles of Counting is used to determine the total number of possible outcomes for a given situation. For instance, if there are a ways of doing something and b ways of doing another thing, there are a total of $a \times b$ ways of doing both actions.

Permutation and Combination

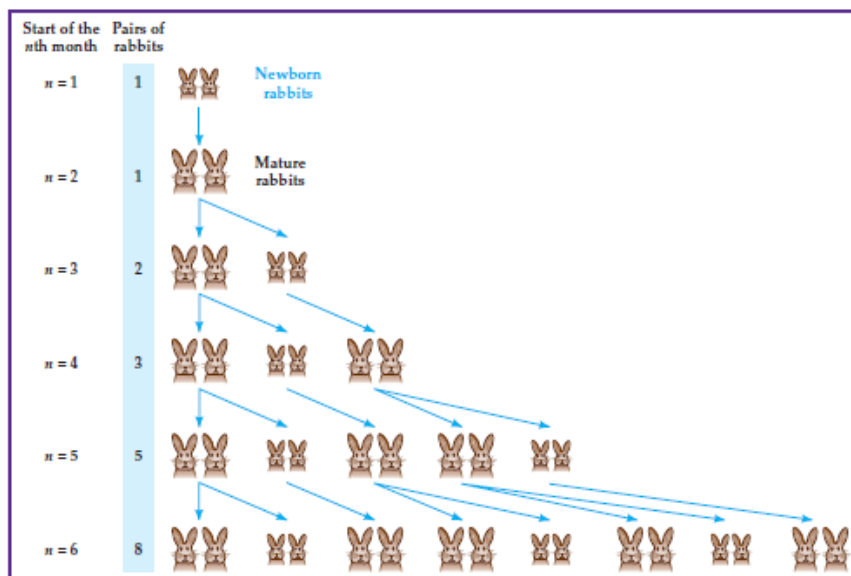
Permutation	Combination
The permutation of n objects taken r at a time determines the number of arrangements where the order <i>is significant</i>	The combination of n objects taken r at a time determines the number of arrangements where the order <i>is not significant</i> .
${}_nP_r = \frac{n!}{(n-r)!}$	${}_nC_r = \frac{n!}{(n-r)!r!}$

V. THE NATURE OF MATHEMATICS: MATHEMATICS IN OUR WORLD

B. THE FIBONACCI SEQUENCE

Leonardo of Pisa, also known as Fibonacci (c. 1170–1250), is one of the best-known mathematicians of medieval Europe. In 1202, after a trip that took him to several Arab and Eastern countries, Fibonacci wrote the book *Liber Abaci*. In this book Fibonacci explained why the Hindu-Arabic numeration system that he had learned about during his travels was a more sophisticated and efficient system than the Roman numeration system. This book also contains a problem created by Fibonacci that concerns the birth rate of rabbits. Here is a statement of Fibonacci's rabbit problem.

At the beginning of a month, you are given a pair of newborn rabbits. After a month the rabbits have produced no offspring; however, every month thereafter, the pair of rabbits produces another pair of rabbits. The offspring reproduce in the same manner. If none of the rabbits dies, how many pairs of rabbits will there be at the start of each succeeding month?



Fibonacci discovered that the number of pairs of rabbits for any month after the first two months can be determined by adding the numbers of pairs of rabbits in each of the two previous months. For instance, the number of pairs of rabbits at the start of the sixth month is $3 + 5 = 8$.

A **recursive definition** for a sequence is one in which each successive term of the sequence is defined using some of the preceding terms. Using F_n to denote the n^{th} Fibonacci number, the numbers in the Fibonacci sequence are given by the following recursive definition:

$$\text{If } F_1 = 1 \text{ and } F_2 = 1, \text{ then } F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 3$$

C. NUMEROUS APPLICATIONS OF MATHEMATICS IN THE WORLD

Fibonacci Numbers in Nature

Fibonacci's rabbit problem is not a very realistic model of the population growth rate of rabbits; however, the numbers in the Fibonacci sequence often occur in nature. For instance, the seeds on a sunflower are arranged in spirals that curve both clockwise and counterclockwise from the center of the sunflower's head to its outside edge. In many sunflowers, the number of clockwise spirals are consecutive Fibonacci numbers. In the sunflower shown at the right, the number of clockwise spirals is 34 and the number of counterclockwise spirals is 55.

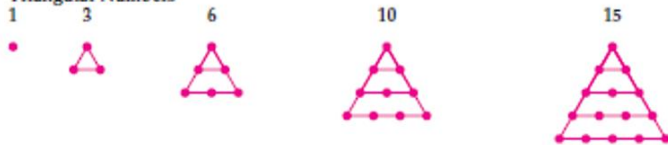
It has been conjectured that the seeds on a sunflower grow in spirals that involve Fibonacci numbers because this arrangement forms a uniform packing. At any stage in the sunflower's development, its seeds are packed so that they are not too crowded in the center and not too sparse at the edge.

Another example can be seen in pineapples, with its spirals known for its hexagonal nubs. The nubs on many pineapples form 8 spirals that rotate diagonally upward to the left and 13 spirals that rotate diagonally upward to the right. Note that 8 and 13 are consecutive Fibonacci numbers.

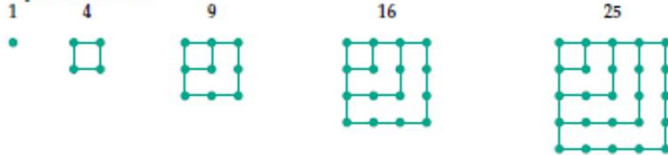


The seeds on this sunflower form 34 clockwise spirals and 55 counterclockwise spirals. The numbers 34 and 55 are consecutive Fibonacci numbers.

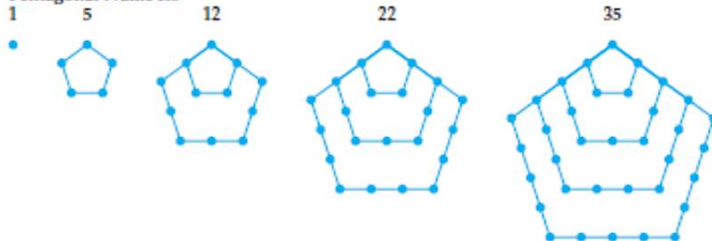
Triangular Numbers



Square Numbers



Pentagonal Numbers



Polygonal Numbers

The ancient Greek mathematicians were interested in the geometric shapes associated with numbers. For instance, they noticed that triangles can be constructed using 1, 3, 6, 10, or 15 dots, as shown below. These numbers are called **triangular numbers**.

Similarly, **square numbers** refer to the numbers 1, 4, 9, 16, 25, ... and **pentagonal numbers** refer to the numbers 1, 5, 12, 22, 35,

An n^{th} -term formula for the triangular numbers is:

$$\text{Triangular}_n = \frac{n(n+1)}{2}$$

The square numbers have an n^{th} -term formula of:

$$\text{Square}_n = n^2$$

The n^{th} -term formula for the pentagonal numbers is:

$$\text{Pentagonal}_n = \frac{n(3n-1)}{2}$$

VI. THE NATURE OF MATHEMATICS: MATHEMATICAL LANGUAGES AND SYMBOLS

A. CHARACTERISTICS OF MATHEMATICAL LANGUAGE: PRECISE, CONCISE, AND POWERFUL

The language of mathematics makes it easy to express the kinds of thoughts that mathematicians like to express. These characteristics are:

1. Precise: *able to make very fine distinctions*
2. Concise: *able to say things briefly*
3. Powerful: *able to express complex thoughts with relative ease*

Characteristics of Mathematical Language

- Mathematical Language is precise, which means it is able to make very fine distinction or definitions among a set of mathematical symbols.

- It is concise because a mathematician can express otherwise long expositions or sentences briefly using the language of mathematics.
- The mathematical language is that is one can express complex thoughts with relative ease. For example, consider the sentence, “The sum of any two real numbers is also a real number.” In mathematical notations, this declarative sentence can be written as $\mathbb{R} + \mathbb{R} = \mathbb{R}$.
- Mathematics is a symbolic language.
- Mathematical language can be used to describe a subset of the real world using only symbols. Problems in physics like freely falling bodies, speed, and acceleration; quantities like the chemical content of the vegetables; the use the mathematical modeling in biological disease modeling; and the formulas employed in the social sciences can be all expressed using mathematical sentences or formulas.
- Mathematical describe abstract structures as well. There are areas of pure mathematics which deal with abstract structures, which have no known physical counterparts at all. These are the studies in areas of mathematics like abstract algebra, linear algebra, topology, real analysis, and complex analysis.
- Mathematics, therefore, is the language of the sciences, business, economics, music, architecture, arts, and even politics. There is an intimate connection between the language of the mathematical and the English language.
- Chinese, Greek, and English language are the same because they communicate ideas through symbols that feed the mind with information. More often however, an idea written in Chinese is written differently in the Greek or English language, resulting in confusion.
- Mathematics tries to avoid this difficult by adopting a universally understood symbolic system for its language. Thus, the language of mathematics can be considered a common language of the world. Any students learning mathematics in all parts of the globe should be able to understand mathematics even if he or she does not understand English.

B. EXPRESSIONS VS SENTENCES

Each language has its vocabulary (the words) and its rules for combining these words into complete thoughts (the sentences). Mathematics is no exception.

As a first step in studying the mathematical language, we will make a very broad classification between the “nouns” of mathematics (used to name mathematical objects of interest) and the “sentences” of mathematics (which state complete mathematical thoughts).

Expressions

An **expression** is the mathematical equivalent to an English noun; it is a correct arrangement of mathematical symbols used to represent a mathematical symbol of interest.

An expression does not state a complete thought; it does make sense to ask if it is true or false, however.

The most common expression types are numbers, sets, and functions. Numbers, for instance, have lots of different names. For instance, the expressions

$$5 \qquad 2 + 3 \qquad \frac{10}{2} \qquad (6 - 2) + 1 \qquad 1 + 1 + 1 + 1 + 1$$

all look different, but are all just different names for the same number. This simple idea—that numbers have lots of different names—is extremely important in Mathematics!

Sentences

A mathematical **sentence** is equivalent to an English sentence; it is a correct arrangement of mathematical symbols that states a complete thought.

Sentences have verbs. In the mathematical sentence $3 + 4 = 7$, the verb is $=$. A sentence can be always true, always false, or sometimes true/false. For instance:

- The sentence $1 + 2 = 3$ is true,
- The sentence $1 + 2 = 4$ is false.
- The sentence $x = 2$ is true or false, depending on the value of x . In this situation, the sentence is true if $x = 2$, and is false otherwise.
- The sentence $x + 3 = 3 + x$ is always true, no matter what number is chosen for x .

Further examples:

Example	Expression or Sentence?
2	Expression
$1 + 1$	Expression
$1 + 1 = 2$	Sentence (true)
$1 + 1 = 3$	Sentence (false)
$x + 1 = 3$	Sentence (sometimes true/false)

C. ELEMENTARY LOGIC, CONNECTIVES, QUANTIFIERS, NEGATION, VARIABLES

Logic Statements and Quantifiers

Logic has been part of our everyday lives. When one uses his/her reasoning skills to make a point or share his/her thoughts, and make a conclusion, logic allows one to reason correctly. The statements that one make are called *propositions*. The formal definition can be seen at the right. Consider the following statements:

1. Tarlac City is in Region II.
2. General Mathematics is one of the subjects in Senior High School.
3. It was raining in Madrid when Magellan arrived in Mactan.
4. How old are you?

In the examples above, it is clear that Sentence (1) is false (Tarlac City is in Region III), and Sentence (2) is true. In both cases, both are propositions. It is unclear in Sentence (3), meanwhile, if it was really raining in Madrid when Magellan arrived in Mactan. However, it is certain that it was either raining in Madrid or not. Thus, Sentence (3) is also a proposition.

Sentence (4), meanwhile is NOT a proposition. Aside from the fact that it is an interrogative sentence, the truth of the statement can only be determined by getting a response to the question.

Propositions are usually denoted by small letters such as $p, q, r, \dots$

Examples:

	Statement	Proposition or not?
p	Alwyn is a Grade 11 student.	The statement is a declarative sentence whose truth value can be known. Therefore, this is a proposition .
q	Ouch!	This is an exclamatory sentence. Therefore, this is not a proposition .
r	$1 + 2 = 3$	This mathematical sentence can be translated to "The sum of 1 and 2 is 3," which is a true declarative sentence. Therefore, this is a proposition .
s	What is a function?	The statement is an interrogative sentence. Therefore, this is not a proposition .
t	If you study hard, then you will get good grades.	This is a declarative statement that can be either true or false. Thus, this is a proposition .
u	If you are in Senior High School, then you are either in Grade 11 or 12.	This is a true declarative sentence. Therefore, this is a proposition .

Truth Tables and Tautologies

Connectives or **logical operators** are used to make compound propositions. Each of these logical operations can be expressed in the following truth table, whose definition can be seen at the right.

Definition
Proposition A proposition is a declarative sentence that is either true or false, but <i>never</i> both.

Definition
Truth Table Given a proposition, its truth table shows all its possible truth values. A truth table involving n proposition has 2^n rows.

Negation	Conjunction	Disjunction																																				
Negation , equivalent to the English word “not”, takes the opposite of the current value.	The conjunction of two propositions p and q , equivalent to the English word “and”, will result to a true value only if both propositions are true.	The disjunction of two propositions p and q , equivalent to the English word “or”, will result to a true value if at least one of the propositions is true.																																				
<table><tr><th>p</th><th>$\sim p$</th></tr><tr><td>T</td><td>F</td></tr><tr><td>F</td><td>T</td></tr></table>	p	$\sim p$	T	F	F	T	<table><tr><th>p</th><th>q</th><th>$p \wedge q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>F</td></tr><tr><td>F</td><td>F</td><td>F</td></tr></table>	p	q	$p \wedge q$	T	T	T	T	F	F	F	T	F	F	F	F	<table><tr><th>p</th><th>q</th><th>$p \vee q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>T</td></tr><tr><td>F</td><td>T</td><td>T</td></tr><tr><td>F</td><td>F</td><td>F</td></tr></table>	p	q	$p \vee q$	T	T	T	T	F	T	F	T	T	F	F	F
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T	F	T																																				
F	T	T																																				
F	F	F																																				
Conditional	Biconditional																																					
The conditional of propositions is equivalent to the English word “implies.” Given $p \rightarrow q$, the proposition p is called the hypothesis , and q is called the conclusion . The conditional is only false when p is true and q is false.	The biconditional of two propositions can be read “if and only if”. Given $p \leftrightarrow q$, p and q are called the components of the biconditional.																																					
<table><tr><th>p</th><th>q</th><th>$p \rightarrow q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td></tr><tr><td>F</td><td>F</td><td>T</td></tr></table>	p	q	$p \rightarrow q$	T	T	T	T	F	F	F	T	T	F	F	T	<table><tr><th>p</th><th>q</th><th>$p \leftrightarrow q$</th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>F</td></tr><tr><td>F</td><td>F</td><td>T</td></tr></table>	p	q	$p \leftrightarrow q$	T	T	T	T	F	F	F	T	F	F	F	T							
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Definition
Simple and Compound Propositions A compound proposition is a proposition formed from simpler propositions using logical connectors (see above) or some combination of logical connectors. If it cannot be broken further in other component propositions, then it is a simple proposition.

From the example above, the only statements that are propositions are p , r , t , and u . From these, p and r are simple propositions and t and u are compound propositions. Among the compound propositions, the logical connectors are the following:

- t : **If** you study hard, **then** you will get good grades.
- u : **If** you are in Senior High School, **then** you are either in Grade 11 **or** 12.

Example:

- Construct a table for $\sim(\sim p \vee q) \vee q$.
- Use the truth table from part a to determine the truth value of $\sim(\sim p \vee q) \vee q$, provided that p is true and q is false.

Solution:

The truth table can be constructed column by column.

p	q	$\sim p$	$\sim p \vee q$	$\sim(\sim p \vee q)$	$\sim(\sim p \vee q) \vee q$
T	T	F	T	F	T
T	F	F	F	T	T
F	T	T	T	F	T
F	F	T	T	F	F

To determine the truth value of $\sim(\sim p \vee q) \vee q$ where p is true and q is false, simply consider the row where the values of p and q are as indicated, which is the row highlighted in green. In this case, the truth value for $\sim(\sim p \vee q) \vee q$ is **true**.

Definitions
Tautology A proposition that is always true is called a tautology , denoted by the Greek letter <i>tau</i> (τ)
Contradiction A proposition that is always false is called a contradiction , denoted by the Greek letter <i>phi</i> (ϕ).
Contingency A proposition that is neither a tautology nor a contradiction is called a contingency .

p	q	r	$\sim p$	$q \wedge r$	$\sim p \vee (q \wedge r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	F	F	F
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	F	T
F	F	F	T	F	T

Example:

Construct the truth table for the compound proposition $\sim p \vee (q \wedge r)$ and determine if it is a tautology, contradiction, or a contingency.

Solution:

Since there are 3 propositions p , q , and r , the truth table will contain $2^3 = 8$ rows. The completed truth table can be seen at the left.

By looking at the rightmost column of the table, it can be seen that there are true values and false values. Thus, the compound proposition $\sim p \vee (q \wedge r)$ is a contingency.

Example:

Construct the truth table for the compound proposition $p \rightarrow (p \vee q)$ and determine whether it is a tautology, contradiction, or a contingency.

Solution:

Since there are two propositions p and q , the truth table will contain $2^2 = 4$ rows. The completed truth table can be seen on the right.

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

By looking at the rightmost column on the table, it can be seen that only true values can be seen. Thus, the compound proposition $p \rightarrow (p \vee q)$ is a tautology.

Forms of Conditional Propositions

Given two propositions p and q , there are three propositions that can be derived from the conditional $p \rightarrow q$:

Form	Description	Logical Expression
Converse	The converse is written by interchanging the hypothesis and conclusion of the conditional statement	Original: $p \rightarrow q$ Converse: $q \rightarrow p$
Inverse	The inverse is written by negating both the hypothesis and conclusion.	Original: $p \rightarrow q$ Inverse: $(\sim p) \rightarrow (\sim q)$
Contrapositive	The contrapositive is done by getting the converse and inverse of the conclusion. In other terms: - Hypothesis and conclusion are interchanged; and - The new hypothesis and conclusion are negated.	Original: $p \rightarrow q$ Contrapositive: $(\sim q) \rightarrow (\sim p)$

Example:

Consider the propositions p : Dylan's strand is ABM and q : Dylan is in the Academic Track. Determine the (a) converse, (b) inverse, and (c) contrapositive of the sentence "If Dylan's strand is ABM, then he is in the Academic Track."

Solution:

The sentence can be written in symbols as $p \rightarrow q$. Thus,

- Converse ($q \rightarrow p$): If *Dylan is in the Academic Track*, then *his strand is ABM*.
- Inverse ($(\sim p) \rightarrow (\sim q)$): If *Dylan's strand is not ABM*, then *he is not in the Academic Track*.
- Contrapositive ($(\sim q) \rightarrow (\sim p)$): If *Dylan is not in the Academic Track*, then *his strand is not ABM*.

VII. THE NATURE OF MATHEMATICS: PROBLEM SOLVING AND REASONING

A. INDUCTIVE AND DEDUCTIVE REASONING

Inductive Reasoning

Inductive reasoning is characterized by the following:

- It is usually based on observation. The premises of inductive arguments are usually bits of evidence that have been gathered by observation, direct or indirect.
- Its conclusions are tentative generalizations about groups, relationships, or predictions.

Some missteps one can make when putting together inductive arguments that can lead to an unsound conclusion include:

- Inferring an unreasonable generalization
- Generalizing from insufficient evidence
- Generalizing from a biased sample

Deductive Reasoning

Deductive reasoning is characterized by the following:

- It generates necessary conclusions. If the premises of the arguments are true—and it is possible that they may not be—then, the conclusion must certainly be true.
- It puts together a general statement about a group and a statement establishing a member of that group and draws a conclusion about that member.
- It puts together a general prediction and a statement about a given situation and draws a conclusion from this. The premises establish what we know about a particular cause and use these to draw conclusions about a situation.

There are two patterns of deductive argument:

1. Syllogisms. This involves putting together a general statement about a group and a statement establishing something as a member of that group and draws a conclusion about that member. See how this is reflected in the logical outline:
 - a. The premises consist of a generalization about a group and a statement about something that belongs to that group.
 - b. The conclusion connects the quality of the group to the thing or person that is part of that group.
 - c. The conclusion must be true if the premises are true.
2. Hypothetical chains. This involves putting together a cause-effect statement (predictions arrived through induction) and a statement about a specific event to draw a conclusion about that event.
 - a. The premises consist of a statement about a cause-effect relationship, another kind of generalization that one would get through induction, and a statement describing a specific situation.
 - b. The conclusion is usually a statement about that specific situation.
 - c. The conclusion must be true if the premises are true.

Deductive reasoning can get wrong when the following is done:

- Syllogisms can go wrong when “All” is considered the same as “Only”.
- Hypothetical chains can go wrong when “Always” is considered the same as “Only”.

C. POLYA'S 4 STEPS IN PROBLEM SOLVING

In 1945 George Polya published the book *How To Solve It* which quickly became his most prized publication. It sold over one million copies and has been translated into 17 languages. In this book he identifies four basic principles of problem solving.

Polya's First Principle: Understand the Problem

- Do you understand all the words used in stating the problem? What are you asked to present or show?
- Can you restate the problem in your own words?
- Can you think of a picture or a diagram that might help you understand the problem?
- Is there enough information to enable you to find a solution?

Polya's Second Principle: Devise a Plan

- Guess and check
- Make an orderly list
- Eliminate possibilities
- Look for a pattern
- Draw a picture
- Solve a simpler problem
- Use symmetry
- Consider special cases
- Use direct reasoning
- Solve an equation
- Use a model
- Work backwards
- Use a formula
- Be ingenious

Polya's Third Principle: Carry Out the Plan

- In carrying out your plan of the solution, check each step.
- Can you see clearly that the step is correct?
- Can you prove that it is correct?

Polya's Fourth Principle: Look Back

- Can you check the result?
- Can you check the argument?
- Can you derive the solution differently?
- Can you see it at a glance?
- Can you see the result, or the method, for some other problem?

VIII. MATHEMATICS AS A TOOL: DATA MANAGEMENT

A. DATA: GATHERING AND ORGANIZING DATA; REPRESENTING DATA USING GRAPHS AND CHARTS

Descriptive Statistics

Descriptive statistics focuses on “describing data in symbolic forms and abbreviated fashions. Sometimes we dealing with a large amount of data and that it is impossible to describe it as it is being a large amount of data but descriptive statistics will provide us certain tools to make the data manageable to handle and conveniently neat to describe.

To explore the characteristics of descriptive statistics, let us create a fictitious situation. What does it mean if someone tells you that majority of workers earn approximately P20,000.00 in a month? Were you able to dissect the idea behind the plain statement? Does it trigger your mind to question further?

This statement is a piece of information that described a particular trait or characteristic of a group of workers. Supplied with this singular information but armed with statistical inquisitiveness, descriptive statistics can further describe the given information to the extent of its depth and breadth.

Inferential Statistics

We could probably argue that descriptive statistics, with its characteristic to describe, is sufficient to depict any given information. While it is effective to describe a manageable size of data, it can hardly engulf a sizeable amount of data. Thus, for this kind of situation, inferential statistics is the alternative technique that can be used. Inferential statistics has the ability to “infer” and to generalize and it offers the right tool to predict values that are not really known.

Let us consider the fictitious situation we made under descriptive statistics, but this time instead of reporting the approximate monthly earning of some workers, we want to determine the estimated monthly earnings of all the workers in a certain region. By attempting to apply descriptive statistics, it would be impossible to ask all the workers in the entire region about their monthly income. But by using inferential statistics, we would instead practically decide to select just a small number of workers and ask them of their monthly income. From there, we can predict or approximate in a “more or less” fashion the monthly income of all workers in the entire region.

Of course, inference or generalization is a risky process that is why we need to ensure that the small group of workers we selected are the approximate representative of the workers in the entire region. But nevertheless, this inference or prediction is better than chance accuracy.

Scales of Measurement

To quantify an observation, it is necessary to identify its **scale of measurement**, also known as **level of measurement**. Scale of measurement is the gateway to the fascinating world of statistics. Without sufficient knowledge of it, all our statistical learnings lead to nowhere.

Marital Status	
Single (1)	Married (2)

1. Nominal Scale. It is concerned with categorical data. It simply means using descriptive categories itself, or using numbers to label categories. This is done by counting the occurrence of frequency within categories. One condition is that the categories must be independent or mutually exclusive.

An example for this is marital status. We can categorize marital status by simply assigning a number. For instance, “1” for single and “2” for married (see table to the right).

Obviously, those numbers only serve as labels and they do not contain any numerical weight. Thus, we cannot say that married people (2) have more marital status than single people (1) on the basis of the numbers assigned to them (i.e.: 2 is greater than 1).

2. **Ordinal Scale.** It is concerned with ranked data. There are instances wherein comparison is necessary and cannot be avoided. Ordinal scale provides ranking of the observation in order to generate information to the extent of “greater than” or “less than”. However, the ranked data generated is limited also to the extent of “greater than” or “less than;”. It is not capable of telling information about how much greater or how much less.

Ordinal scale can be best illustrated in sports activities like fun run. Finding the order of finish among the participants in a fun run always come up with a ranking. However, ranked data cannot provide information as to the difference in time between 1st placer and 2nd placer. Relative to this, reading reports with ordinal information is also tricky. For example, a TV commercial extol a certain brand for being the number one product in the country. This may seem acceptable, but if you learned that there is no other product then definitely the message of the commercial will be swallowed with a smirking face.

3. **Interval Scale.** It deals with measurement of data. In the nominal scale, we use numbers to label categories while in the ordinal scale we use numbers to merely provide information regarding greater than or less than. However, in interval scale we assign numbers in such a way that there is meaning and weight on the value of points between intervals. This scale of measurement provides more information about the data. Consider the comparative illustration below:

Academic performance of five students in a certain class

	Student A	Student B	Student C	Student D	Student E
Interval Data	99	74	73	70	70
Ordinal Data	1 st	2 nd	3 rd	4 th	5 th
Nominal Data	Passed	Failed	Failed	Failed	Failed

As what can be noticed, the interval scale provides substantial information about the grades of students compared to the earlier two. Student A earned a grade of 99, and so on and so forth. Now look at the information given by ordinal data. It is simply about ranking. With this of information, Student B can proudly and rightfully claim the 2nd place in the ranking. Ordinal scale is a trusted friend to keep a secret, that the grade of student B, though claiming 2nd place, is actually 74.

Analyzing the nominal data in the example, it can be seen that only one student passed. Nominal data cannot provide more information specifically provide brighter limelight to student A. Audience may assume that Student A just got passing grade a little bit higher than the passing mark but student A grade of 99 will remain hidden forever.

4. **Ratio Scale.** This is an extension of the interval scale. Similar to the interval scale, it also deals with measurement of data, but the difference between the two is that the ratio scale involves the concept of *absolute value*. Because of this, we oftentimes cannot utilize ratio scale in the social sciences. We cannot justify an absolute value to gauge intelligence. We cannot say that our student A with a grade of 99 has an intelligence several points superior than student E who hardly but successfully achieved a grade of 70.

Key Concepts in Statistics

Concept	Description
Population	A population can be defined as an entire group of people, things, or events having at least one trait in common. This common trait is the binding factor in order to group a cluster and call it a population.
Parameter	In gauging the entire population, any measure obtained is called a parameter . Situationally, if someone asks you as to what is the parameter of the study, then bear in mind that he is referring to the size of the entire population. In some situations where the actual size of the population is difficult to obtain; the parameters are in the form of estimate or inference.
Sample	The small number of observations taken from the total number making up a population is called a sample . As long as the observation or data is not the totality of the entire population, then it is always considered a sample. For instance, in a population of 100, then 1 is considered as a sample. 30 is clearly a sample. It may seem absurd but 99 taken from 100 is still considered a sample. Not until we include that last number (making it 100) could we claim that it is already a population and no longer a sample.
Statistic	In gauging the sample, any measure obtained from the sample is called a statistic . Since a sample is easier to gather or observe than the population, then statistics are simpler to gather than parameters.

B. INTERPRETING ORGANIZED DATA

Graphing is one way of visually showing the behavior of data. To create a graph, the distribution of scores must be organized. For instance, the following scores arranged in an unorganized manner can provide confusing or no information at all.

120 65 110 75 105 80 105 85 100 85 100 90 95 90 90

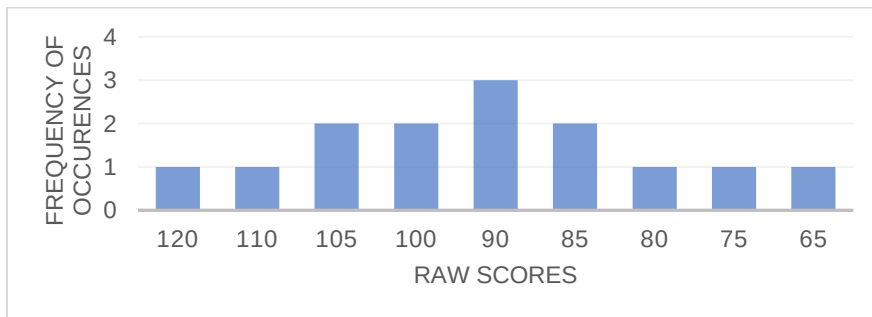
However, when these scores are arranged from highest to lowest, for instance, a **score distribution** is formed where some pieces of information can be brought forth and exposed.

120 110 105 105 100 100 95 90 90 90 85 85 80 75 65

The score distribution can still be organized in a form of a frequency distribution (see right). **Frequency distribution** provides information about raw scores, and the frequency of occurrences. Frequency distribution provides clearer insights about the behavior of scores.

Another alternative way of presenting data in frequency distribution is to present them in a tabular form. A tabular form has the advantage of showing the visual representation of the data. This kind of presentation is more appealing to the general audience.

x (Raw Score)	f (Frequency of Occurrence)
120	1
110	1
105	2
100	2
95	1
90	3
85	2
80	1
75	1
65	1



Another way of showing data in graphical form is the frequency polygon (see below).



C. MEASURES OF CENTRAL TENDENCY (GROUPED AND UNGROUPED DATA)

Ungrouped Data

1. **Mean.** The mean (also known as the arithmetic mean) is the most commonly used measure of central position. It is used to describe a set of data where the measures cluster or concentrate at a point. As the measures cluster around each other, a single value appears to represent distinctively the typical value. It is the sum of all measures x divided by the number N of measures in a variable. It is symbolized as $\bar{x}$ (read as "x bar"). In symbols,

$$\bar{x} = \frac{\sum x}{N}$$

where $\sum x$ is the summation of x (sum of the measures) and N is the number of values of x .

Example: The grades of 10 students in Geometry are 87, 84, 85, 85, 86, 90, 79, 82, 78, and 76. What is the average grade of the 10 students?

Solution:

$$\begin{aligned}\bar{x} &= \frac{\sum x}{N} \\ \bar{x} &= \frac{87 + 84 + 85 + 85 + 86 + 90 + 79 + 82 + 78 + 76}{10} \\ \bar{x} &= \frac{832}{10} \\ \bar{x} &= 83.2\end{aligned}$$

Hence, the average grade of the 10 students is 83.2.

2. **Median.** The median is the middle value or term in a set of data arranged according to size/magnitude (either increasing or decreasing).

Example: The library logbook shows that 58, 60, 54, 35, and 97 books were borrowed from Monday to Friday, respectively. Find the median.

Solution:

Arrange the data in increasing order: 35, 54, 58, 60, 97.

It can be seen from the arranged numbers that the middle number is 58. Therefore, the median is 58.

Example: Andrea's score in 10 quizzes during the first quarter are 8, 7, 6, 10, 9, 5, 9, 6, 10, and 7. Find the median.

Solution:

Arrange the data in increasing order: 5, 6, 6, 7, 7, 8, 9, 9, 10, 10.

Since the number of measures is even, there are two "middlemost" scores: 7 and 8. The median is determined by computing the median of the two: $\frac{7+8}{2} = 7.5$. Thus, the median is 7.5.

3. **Mode.** The mode is the measure or value which occurs most frequently in a set of data. It is the value with the greatest frequency. To find the mode for the set of data:

- Select the measure that appears most often in the set.
- If two or more measures appear the same number of times, then each of these values is a mode.
- If every measure appears the same number of times, then the set of data has no mode.

Example: Find the mode of the following set of data:

- a. 10, 12, 9, 10, 13, 11, 10
- b. 15, 20, 18, 19, 18, 16, 20, 18

Solution:

- a. In the given set of data, 10 appears the most often. Thus, the mode is 10.
- b. In the given set of data, 18 appears the most often. Thus, the mode is 18.

Grouped Data

1. **Mean.** When the number of items is too big, then items are grouped for convenience. To find the mean of grouped data using class marks, the following formula can be used:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

where f is the frequency of each class and x is the class mark.

Example: Calculate the mean of the Mid-year Test scores of students in Filipino.

Score	Frequency
41 – 45	1
36 – 40	8
31 – 35	8
26 – 30	14
21 – 25	7
16 – 20	2

Solution:

To solve, we need to add two more columns: one for the class mark $x = \frac{LL+UL}{2}$ and another to get the product of f and x .

Score	Frequency f	Class mark x	fx
41 – 45	1	43	43
36 – 40	8	38	304
31 – 35	8	33	264
26 – 30	14	28	392
21 – 25	7	23	161
16 – 20	2	18	36
$i = 5$	$\sum f = 40$		$\sum fx = 1200$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1200}{40} = 30$$

Therefore, the mean of the mid-year test scores is 30.

2. **Median.** The median is the middle value in a set of quantities. It separates an ordered set of data into two equal parts. Half of the quantities is located above the median and the other half is found below it, whenever the quantities are arranged according to magnitude (from highest to lowest). To solve for the median, the following formula is used:

$$\tilde{x} = lb_{mc} + \left(\frac{\frac{\sum f}{2} - <cf}{f_{mc}} \right) i$$

where: lb_{mc} is the lower boundary of the median class;

f is the frequency of each class;

$<cf$ is the cumulative frequency of the lower class next to the median class;

f_{mc} is the frequency of the median class; and

i is the class interval

The median class is the class interval that contains the $\frac{\sum f}{2}$ quantity. The computed median must be within the median class.

Example: Calculate the median of the Mid-year Test scores of students in Filipino.

Score	Frequency
41 – 45	1
36 – 40	8
31 – 35	8
26 – 30	14
21 – 25	7
16 – 20	2

Solution:

To solve, we need to add two more columns: one for the class mark $x = \frac{LL+UL}{2}$ and another to get the product of f and x .

Score	Frequency f	lb	$<cf$
41 – 45	1	40.5	40
36 – 40	8	35.5	39
31 – 35	8	30.5	31
26 – 30	14	25.5	23
21 – 25	7	20.5	9

Median class

16 – 20	2	15.5	2
$i = 5$	$\Sigma f = 40$		

To find the median, the following has to be determined:

- To determine the median class, $\frac{\Sigma f}{2} = \frac{40}{2} = 20$. Since the 20th score is within the interval 26 – 30, that is the median class.
- $< cf = 9$
- $f_{mc} = 14$
- $lb_{mc} = 25.5$
- $i = 5$

Thus,

$$\tilde{x} = lb_{mc} + \left(\frac{\frac{\Sigma f}{2} - < cf}{f_{mc}} \right) i$$

$$\tilde{x} = 25.5 + \left(\frac{20 - 9}{14} \right) 5$$

$$\tilde{x} = 25.88$$

Thus, the median of the mid-year test scores is 25.88.

3. Mode. The mode of grouped data can be approximated using the following formula:

$$\hat{x} = lb_{mc} + \left(\frac{D_1}{D_1 + D_2} \right) i$$

where: lb_{mc} is the lower boundary of the modal class;

D_1 is the difference between the frequencies of the modal class and the next upper class;

D_2 is the difference between the frequencies of the modal class and the next lower class; and

i is the class interval.

The modal class is the class with the highest frequency.

Example: Calculate the mode of the Mid-year Test scores of students in Filipino.

Score	Frequency
41 – 45	1
36 – 40	8
31 – 35	8
26 – 30	14
21 – 25	7
16 – 20	2

Solution:

To solve, we need to add two more columns: one for the class mark $x = \frac{LL+UL}{2}$ and another to get the product of f and x .

Score	Frequency f	lb
41 – 45	1	40.5
36 – 40	8	35.5
31 – 35	8	30.5
26 – 30	14	25.5
21 – 25	7	20.5
16 – 20	2	15.5
$i = 5$	$\Sigma f = 40$	

Modal class

To find the median, the following has to be determined:

- The modal class is the class interval with the highest frequency. Since 14 has the highest frequency, the modal class is 26 – 30.
- $lb_{mc} = 25.5$
- $D_1 = 14 - 8 = 6$
- $D_2 = 14 - 7 = 7$

$$e. \quad i = 5$$

Thus,

$$\hat{x} = lb_{mc} + \left(\frac{D_1}{D_1 + D_2} \right) i$$

$$\hat{x} = 25.5 + \left(\frac{6}{6 + 7} \right) 5$$

$$\hat{x} = 28.19$$

Thus, the mode of the mid-year test scores is 28.19.

D. MEASURE OF VARIABILITY (UNGROUPEd DATA)

Variability measures how spread out or dispersed a data set is. It describes how the data sets vary by comparing them to other sets of data.

1. **Range.** The range, denoted by R , is a measure of variability that describes the scores' variability by providing the width of the distribution. The range can be obtained by simply getting the difference between the highest and lowest scores.

$$R = \text{highest score} - \text{lowest score}$$

The range is easy to solve; however, it is very much affected by outliers. It is not stable enough to indicate variability.

2. **Standard Deviation.** The standard deviation is another measure of variability and it is known as the life-blood of the variability concept. It provides measurement about how much all of the scores in the distribution differ from the mean. It describes how clustered or spread out your data around the mean. Unlike the range, the standard deviation is not easily affected by outliers. Thus, it is known to be the most stable measures of variability.

The standard deviation can be computed as the following:

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum(x^2 - \mu)}{N}}$$

where:

x is the raw score in a distribution;
 μ is the population mean; and
 N is the population size.

Sample Standard Deviation

$$s = \sqrt{\frac{\sum x^2 - \bar{x}}{n - 1}}$$

where:

x is the raw score in a distribution;
 $\bar{x}$ is the sample mean; and
 n is the sample size

A distribution with small standard deviation indicates that the data set are more clustered and that the scores are homogenous, while a distribution with a large standard deviation indicates that the data set are more spread apart and that the scores are heterogenous. In addition, a zero standard deviation implies that the scores are of the same value.

3. **Variance.** Variance is also a measure of variability which implies the square of the standard deviation. It calculates the average degree by which each score differs from the mean of the distribution.

The variance is computed as:

Population Standard Deviation

$$\sigma^2 = \frac{\sum(x^2 - \mu)}{N}$$

where:

x is the raw score in a distribution;
 μ is the population mean; and
 N is the population size.

Sample Standard Deviation

$$s^2 = \frac{\sum x^2 - \bar{x}}{n - 1}$$

where:

x is the raw score in a distribution;
 $\bar{x}$ is the sample mean; and
 n is the sample size

F. MEASURES OF RELATIVE POSITION

1. **z-score**. The z-score (also known as the standard score) measures how many standard deviations in an observation is above or below the mean. A positive z-score measures the number of standard deviations a score is above the mean, and a negative z-score mean the number of standard deviations a score is below the mean.

The following is how the z-score is computed depending on whether the population or the sample is involved:

Population z-score

$$z_x = \frac{x - \mu}{\sigma}$$

Sample z-score

$$z_x = \frac{x - \bar{x}}{s}$$

As the median divides the set of scores into two equal parts, there are other measures that divide the distribution into one hundred, four, or ten equal parts. These are the other measures of position: the percentiles, the quartiles, and the deciles.

2. **Percentiles**. One way of assessing performance s by the use of percent. The percentiles are the score-points that divide a distribution into 100 equal parts.

Definition
p^{th} Percentile A value x is called the p^{th} percentile of a data set, provided $p\%$ of the data values are less than x

Meanwhile, the **percentile rank** determines how many percent of the cases got below the rank or position.

Definition
Percentile rank Given a set of data and a data value x , $\text{Percentage rank of score } x = \frac{\text{number of data values less than } x}{\text{total number of data values}} \times 100$

3. **Quartiles**

The quartiles are points that divide a distribution into four equal parts. The quartiles Q_1 is called the first quartile, Q_2 is the second quartile, and so on.

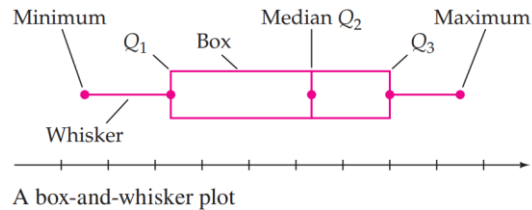
Median Procedure in Finding Quartiles
1. Rank the data. 2. Find the median of the data. This is the second quartile Q_2. 3. The first quartile Q_1 is the median of the data values less than Q_2. The third quartile Q_3 is the median of the data values greater than Q_2.

4. **Deciles**

Deciles are points that divide a distribution into ten equal parts.

Box-and-Whisker Plots

A box-and-whisker plot (sometimes called a boxplot) is a graph that presents information from a five-number summary.



Constructing a Box-and-Whisker Plot

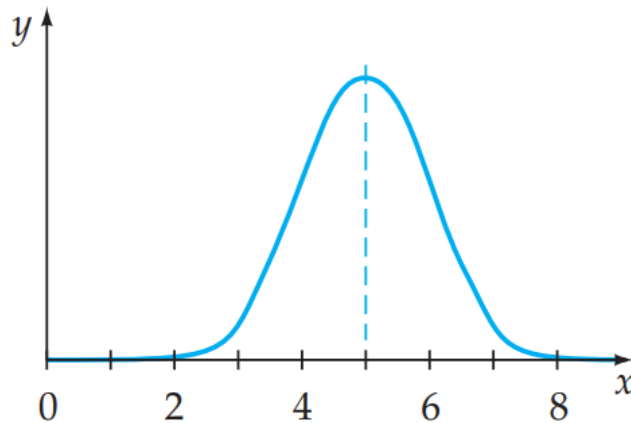
1. Draw a horizontal scale that extends from the minimum data value to the maximum data value.
2. Above the scale, draw a rectangle (box) with its left side at Q_1 and its right side at Q_3 .
3. Draw a vertical line segment across the rectangle at the median Q_2 .
4. Draw a horizontal line segment, called a whisker, that extends from Q_1 to the minimum and another whisker that extends from Q_3 to the maximum.

G. PROBABILITIES AND NORMAL DISTRIBUTIONS

Normal Distribution

A normal distribution, sometimes called the bell curve, is a distribution that occurs naturally in many situations. It is symmetric about a vertical line through the mean of the data.

1. In statistics, the normal distribution is called the normal curve.
2. In the social sciences, it is called the bell curve because of its shape.
3. In physics, it is called the Gaussian distribution.



Every normal distribution has the following properties:

1. The graph is symmetric about a vertical line through the mean of the distribution.
2. The mean, median, and mode are equal.
3. The y -value of each point on the curve is the *percent* (expressed as a decimal) of the data at the corresponding x -value.
4. Areas under the curve that are symmetric about the mean are equal.
5. The total area under the curve is 1.

Standard Normal Distribution

This is a distribution of a normal random variable with mean equal to 0 and standard deviation equal to 1.

H. LINEAR REGRESSION AND CORRELATION

Correlation

Correlation refers to the statistical association between two variables. A correlation exists between two variables when one of them is related to the other in some way.

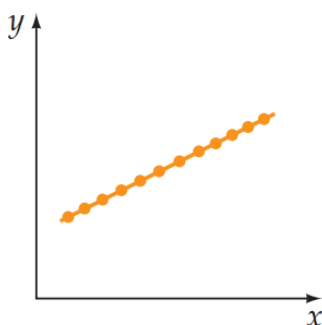
A scatterplot is the best place to start. A scatterplot (also called a scatter diagram) is a graph of a paired (x, y) sample data with a horizontal x -axis and a vertical y -axis. Each individual (x, y) pair is plotted as a single point. A scatterplot can identify several different types of relationships between two variables.

1. A relationship has no correlation when the points on a scatterplot do not show any direction or pattern.
2. A relationship is non-linear when the points on a scatterplot follow a pattern but not a straight line.
3. A relationship is linear when the points on a scatterplot follow a somewhat straight line pattern.

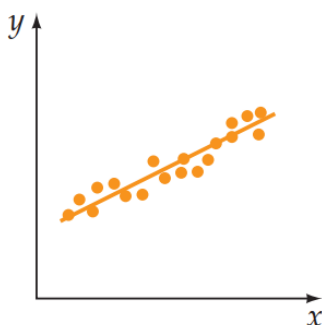
Linear Correlation Coefficient

To determine the strength of a linear relationship between two variable, statisticians use a statistic called the linear correlation coefficient, denoted as r and is defined as follows:

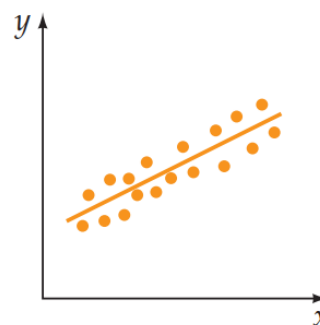
Linear Correlation Coefficient
For the n ordered pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the linear correlation coefficient r is given by
$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}}$



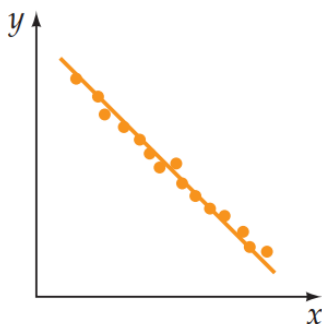
a. Perfect positive correlation, $r = 1$



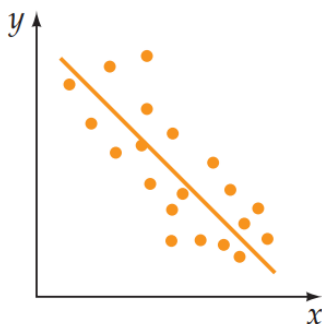
b. Strong positive correlation, $r \approx 0.8$



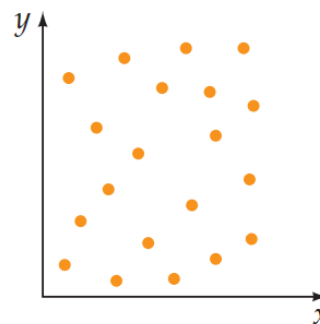
c. Positive correlation, $r \approx 0.6$



d. Strong negative correlation, $r \approx -0.9$



e. Negative correlation, $r \approx -0.5$



f. Little or no linear correlation, $r \approx 0$

Linear Regression

A linear regression model is a mathematical equation that allows us to predict a response for a given predictor value. This is used in the process of prediction. Prediction is calculating a score of the criterion variable $\hat{y}$ on the basis of the knowledge of the predictor x .

Linear Regression

Linear regression can be computer using the equation

$$\hat{y} = a + bx$$

where:

a is the y -intercept

b is the slope

x is the predictor value; and

$\hat{y}$ is the estimate of the mean value of the response variable for any value of the predictor variable

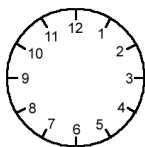
This is also called the least squares regression line.

IX. MATHEMATICS AS A TOOL: MATHEMATICAL SYSTEMS

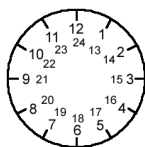
A. MODULAR ARITHMETIC

Most people are familiar with the idea that $1 + 1 = 2$ and will not accept any other answers other than 2. What most people do not recognize is their familiarity with **modular arithmetic**, also known as clock arithmetic. Modular arithmetic teaches us addition and multiplication *modulo* n . As a result, unusual ideas like $1 + 1 = 0$ and $3 \times 2 = 0$. These new concepts will help us see that is fine to think what will happen if certain conditions are changed.

Modular arithmetic features the power of remainders in problem solving. One way to introduce the concept of modular arithmetic is through the ordinary clock.



The numbers of the clock go from 1 to 12.



If we keep on going around the clock, when we get to “25 o’clock”, we are actually back to where 1 o’clock is on the clock face of the clock.

From the example above, we have discussed that $13 = 1 + \text{some multiple of } 12$, and $25 = 1 + \text{some multiple of } 12$, or in other words, “the remainder when 13 is divided by 12 is 1” and “the remainder when 25 is divided by 12 is 1”, respectively.

These two are written mathematically as $13 \equiv 1 \pmod{12}$ (read “13 is congruent to 1 modulo 12”) and $25 \equiv 1 \pmod{12}$ (read “25 is congruent to 1 modulo 12”), respectively.

Modular Addition and Multiplication

In arithmetic modulo n , we take the answer *mod* n when adding, subtracting, or multiplying two numbers. For instance, when getting the product of two numbers *modulo* n , then multiply the two numbers as normal and the answer is the remainder when the normal product is divided by n . The value n is sometimes called the **modulus**.

Specifically, let $\mathbb{Z}_n$ represent the set $\{0, 1, 2, \dots, n - 1\}$ and define the two operations:

$$a+_nb = (a + b) \bmod n$$

$$a\cdot_nb = (a \times b) \bmod n$$

Examples:

$31 \equiv 7 \pmod{10}$	$31 - 7 = 24$, and 24 is a multiple of 10
$43 \equiv 1 \pmod{7}$	$43 - 1 = 42$, and 42 is a multiple of 7

Some further examples:

Modular Addition	Modular Multiplication
$4 = 2 = 6 \equiv 1 \pmod{5}$	$5 \times 6 = 30 \equiv 2 \pmod{7}$
$3 + 6 = 9 \equiv 2 \pmod{7}$	$4 \times 7 = 28 \equiv 13 \pmod{15}$

Modular arithmetic can also be expressed as tables. The following are examples of modular addition and multiplication for modulo 4 and modulo 5.

Modulo	Modular Addition	Modular Multiplication
Modulo 4	$\begin{array}{c cccc} +_4 & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 1 & 2 & 3 \\ 1 & 1 & 2 & 3 & 0 \\ 2 & 2 & 3 & 0 & 1 \\ 3 & 3 & 0 & 1 & 2 \end{array}$	$\begin{array}{c cccc} \cdot_4 & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 & 3 \\ 3 & 0 & 3 & 2 & 1 \end{array}$
Modulo 5	$\begin{array}{c ccccc} +_5 & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 1 & 2 & 3 & 4 \\ 1 & 1 & 2 & 3 & 4 & 0 \\ 2 & 2 & 3 & 4 & 0 & 1 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 4 & 4 & 0 & 1 & 2 & 3 \end{array}$	$\begin{array}{c ccccc} \cdot_5 & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 3 & 4 \\ 2 & 0 & 2 & 4 & 1 & 3 \\ 3 & 0 & 3 & 1 & 4 & 2 \\ 4 & 0 & 4 & 3 & 2 & 1 \end{array}$

B. APPLICATIONS

Modular Arithmetic in Time

One intuitive use of the modular arithmetic is with a clock. There are only 12 hours on the face of an ordinary clock. If it is 11:00 now, then in 5 hours the clock will show 4:00 instead of 16:00. 4 is the remainder of 16 with a modulus of 12. Another example, if the time now is 7 o'clock, 20 hours later will be 3 o'clock and we do not say 27 o'clock. This is the reason why modular arithmetic is also referred as **clock arithmetic**.

Now, let us have another example of modular arithmetic application to time:



It is currently 7:00 PM. In 1000 hours, what time (in AM/PM) will it be?

Solution: Since time repeats every 24 hours, modulo 24 will be used.

Since $1000 = 16 + (24 \times 41) \equiv 16 \pmod{24}$, the time in 1000 hours is equivalent to 16:00, which is equivalent to 4:00 PM.

As can be seen from the example above, modular arithmetic can be used in simple time calculations. To generalize:

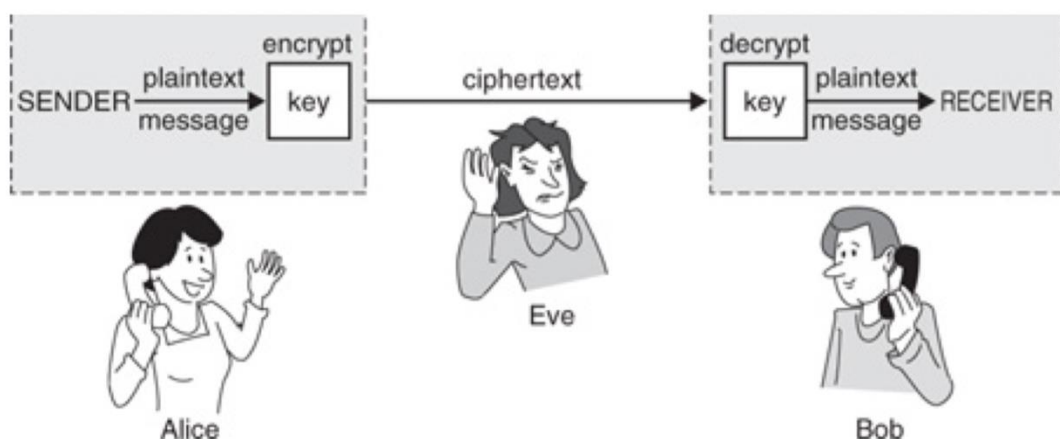
If the current time is t , and the time n hours in the future is f , the future time is computed depending on whether a 12-hour or a 24-hour clock is used:

12-hour Clock	24-hour Clock
$t + n = f \pmod{12}$	$t + n = f \pmod{24}$

Mathematics behind Cryptology

Cryptology is the study of secrecy systems, which consists of two parts: cryptography and cryptanalysis. One of the concepts of number theory used in cryptology is modular arithmetic.

Cryptography is the practice and study of techniques for secure communication in the presence of adverse third parties.



The Caesar Cipher

The algorithm of the Caesar Cipher is to replace each letter by the k^{th} letter preceding it in the alphabet. The specific choice of k in a given instance of this cypher is the *key*. Below is the table of letters with their respective numbers accordingly.

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

In general, fix an alphabet Σ and let $m = |\Sigma|$. Then, fix a secret key, an integer k such that $0 < k < m$. The encryption and decryption functions are:

$$e_k(x) \equiv (x + k) \bmod m$$

$$d_k(y) \equiv (y - k) \bmod m$$

Example:

Consider $\Sigma = \{A, B, C, \dots, Z\}$, and so, $m = 26$. Encrypt "HELP" using the Caesar Cipher with $k = 7$.

Encryption	Decryption
$H \rightarrow 7 \Rightarrow e(7) = (7 + 7) \bmod 26 \equiv 14 \rightarrow O$ $E \rightarrow 4 \Rightarrow e(4) = (4 + 7) \bmod 26 \equiv 11 \rightarrow L$ $L \rightarrow 11 \Rightarrow e(11) = (11 + 7) \bmod 26 \equiv 18 \rightarrow S$ $P \rightarrow 15 \Rightarrow e(15) = (15 + 7) \bmod 26 \equiv 22 \rightarrow W$	To decrypt "OLSW", the decryption function is used: $O \rightarrow 14 \Rightarrow d(14) = (14 - 7) \bmod 26 \equiv 7 \rightarrow H$ $L \rightarrow 11 \Rightarrow d(11) = (11 - 7) \bmod 26 \equiv 4 \rightarrow E$ $S \rightarrow 18 \Rightarrow d(18) = (18 - 7) \bmod 26 \equiv 11 \rightarrow L$ $W \rightarrow 22 \Rightarrow d(22) = (22 - 7) \bmod 26 \equiv 15 \rightarrow P$

Thus, the encrypted word is "OLSW".

X. APPORTIONMENT AND VOTING

A. INTRODUCTION TO APPORTIONMENT

The mathematical investigation into **apportionment**, which is a method of dividing a fixed number of items among groups of different sizes.

Since 1790, when the United States House of Representatives first attempted to apportion itself, various methods have been used to decide how many voters would be represented by each member of the House. The two competing plans were put forward by Alexander Hamilton and Thomas Jefferson.

The Hamilton Plan

Under the Hamilton plan, the total population of the country is divided by the number of representatives. This gives the ***standard divisor***, the number of citizens represented by each representative.

Today, apportionment is applied to situations other than a population of people. For instance, *population* can refer to the number of math classes offered at a college, or the number of fire stations in a city. Nonetheless, the definition of *standard divisor* is still given as though people were involved.

The Jefferson Plan

As can be seen with the Hamilton plan, dividing by the standard divisor and then rounding down does not always yield the correct number of representatives. To overcome this difficulty, the Jefferson plan uses a *modified standard divisor*. This number is chosen by trial and error so that the sum of the standard quotas is equal to the total number of representatives. In a specific apportionment calculation, there may be more than one number that can serve as the modified standard divisor.

Definitions
Standard Divisor $\text{standard divisor} = \frac{\text{total population}}{\text{number of people to apportion}}$
Standard Quota The standard quota is the whole number part of the quotient of a population divided by the standard divisor.